

RA8T2 Group

CN032-9 DC Servo Solution Startup User Guide

Introduction

This startup guide is a quick start guide for CN032-9 DC Servo Solution by using workbench and TwinCAT.

Target Device

RA8T2 Group

When applying the sample program covered in this startup guide to another microcomputer, modify the program according to the specifications for the target microcomputer and conduct an extensive evaluation of the modified program.

Related Document

CN032-9 DC Servo Solution - Encoder Board Software Manual

CN032-9 DC Servo Solution - Encoder Board Hardware Manual

CN032-9 DC Servo Solution - Main Control Board Software Manual

CN032-9 DC Servo Solution - Main Control Board Hardware Manual

CN032-9 DC Servo Solution - Startup Manual (this manual)

Contents

1. Overview	3
1.1 CN032-9 DC Servo Solution Overview	3
1.2 Connection Configuration	3
1.3 Connection Equipment	4
1.3.1 24V CN032-9 DC Servo Solution Main Control Board Motor	4
1.3.2 Absolute Encoder	4
1.3.3 Communication with the PC	5
1.3.3.1 Motor Workbench Communication Board	5
1.3.3.2 RICE	6
1.3.4 Power Supply	7
2. Operating Environment	8
3. CN032-9 DC Servo Solution Execution Procedure	9
3.1 DC Servo Solution Starting	9
3.2 Renesas Motor Workbench Execution	10
3.2.1 Installation	10
3.2.2 Demonstration Mode	10
3.2.2.1 Speed Control Mode	10
3.2.2.2 Position Control Mode	11
3.3 Renesas Motor Workbench Execution	12
3.3.1 Preparing EtherCAT Communication	12
3.3.2 TwinCAT®3 Installation	12
3.3.3 Copying the ESI File	16
3.3.4 Confirm Communication with TwinCAT®3	16
3.3.4.1 Starting TwinCAT®3	16
3.3.4.2 Checking the Communication Status	20
3.3.5 CiA402 Drive Profile Operation	22
3.3.5.1 Operation mode setting	22
3.3.5.2 Cyclic Synchronous Position Mode (csp)	23
3.3.5.3 Cyclic Synchronous Velocity Mode (csv)	25
3.3.6 EEPROM Data Update on CN032-9 DC Servo Solution	28
4. Reference Documents	31
Revision History	Error! Bookmark not defined.

1. Overview

1.1 CN032-9 DC Servo Solution Overview

This application note is implemented by using the board equipped with Renesas Electronics RA8T2 and the Inverter circuit for 24V DC Servo Solution motor.

This solution includes software that runs on the solution board, Renesas motor workbench in PC, and circuit diagram. Therefore, initial evaluation and advance development of industrial motor equipment development using RA8T2 can be easily done.

1.2 Connection Configuration

Figure 1-1 is shown the connection configuration.

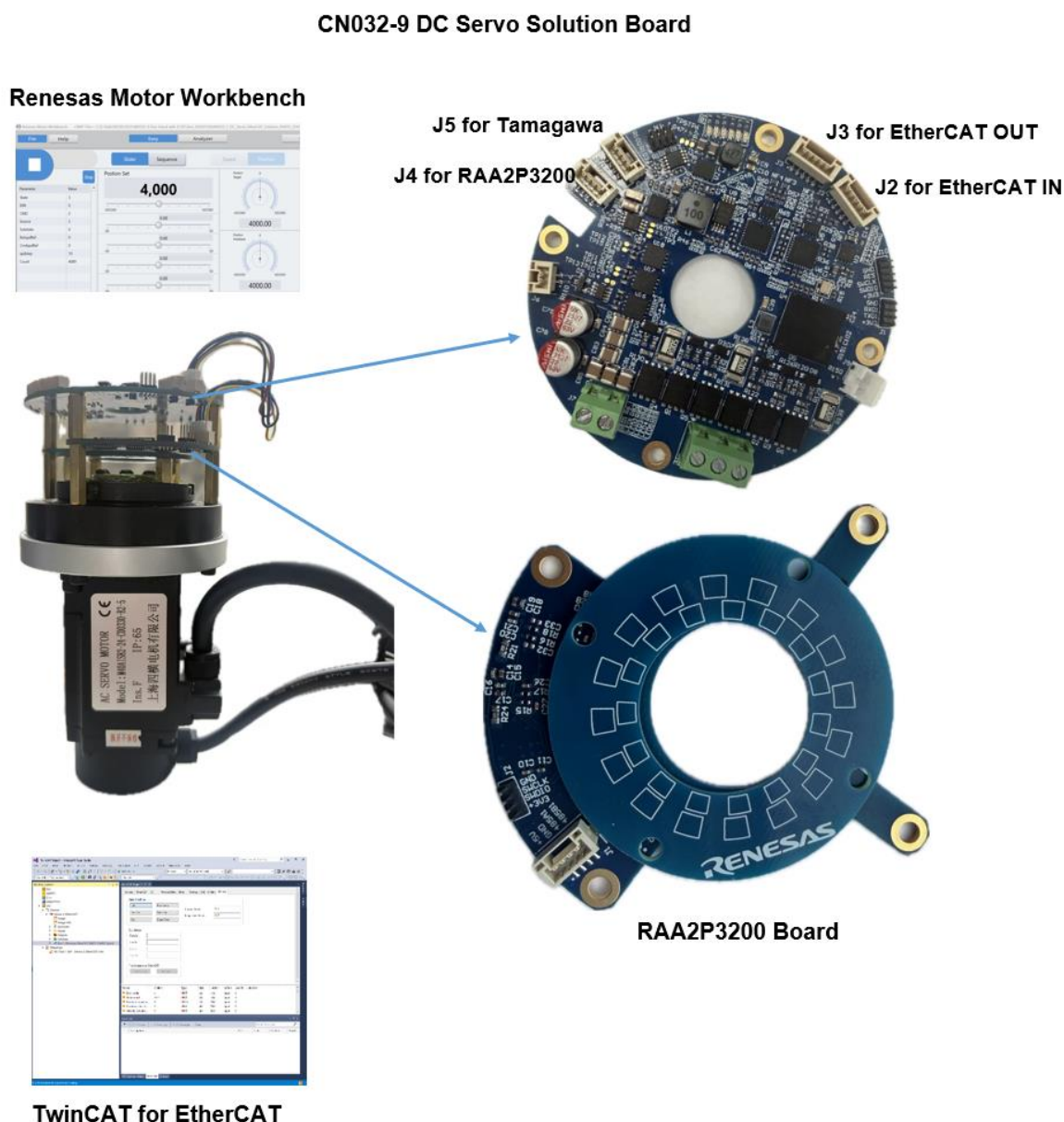


Figure 1-1 Connection Configuration

1.3 Connection Equipment

Table 1-1 is shown the connection equipment list.

Table 1-1 Connection Equipment List

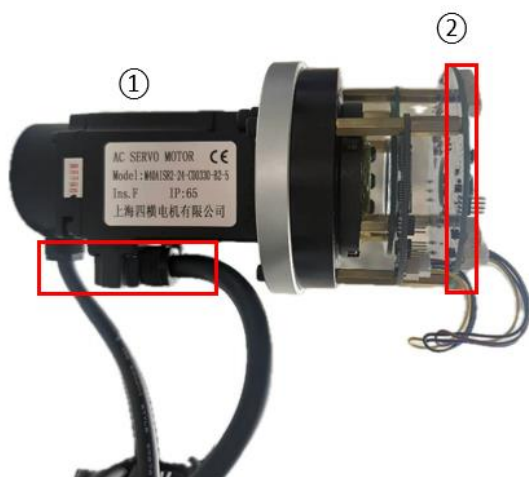
No.	Item	Model	Spec.
1	24V DC servo motor	M40AIS82-24-C00330-B2-5 ZhuHai KaiBang Motor Manufacturing Co. ,	156W, 3000rpm
2	Encoder1	TS5669N140 Tamagawa Seiki Co., Ltd.	17Bit
3	Encoder2	RAA2P3200	12Bit ^{Note}
4	Communication with the PC	Motor workbench communication board	-
		Ethernet interface	-
5	ICE	- J-Link EDU from Segger	-

Note : The customer can enhance the existing mechanical structure to achieve higher accuracy.

1.3.1 24V CN032-9 DC Servo Solution Main Control Board Motor



1.3.2 Absolute Encoder



① TS5669N140

② RAA2P3200

1.3.3 Communication with the PC

1.3.3.1 Motor Workbench Communication Board

The communication board can communicate data between CN032-9 DC servo solution main control board and Renesas motor workbench in PC.

Table 1-2 is shown the connection with CN032-9 DC servo solution main control board, communication board and PC.

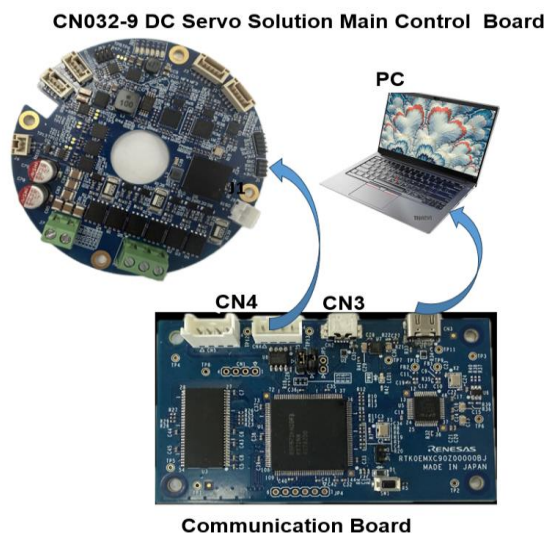


Figure 1-2 Using communication board to connect to host PC

Table 1-2 Connection communication board and RA8T2 motor board

Pin Number	CN4 in communication board	J1 in RA8T2	Note
1	3.3V	3.3V	Power supply
2	TXD from MCU	TXD	Communication interface
3	RXD from MCU	RXD	Communication interface
4	GND	GND	Ground

The Ethernet interface board can communicate data between CN032-9 DC servo solution main control board and TwinCAT in PC.

Figure 1-3 is shown the connection with CN032-9 DC servo solution main control board, Ethernet interface board and PC.

CN032-9 DC Servo Solution Main Control Board

Ethernet Interface Board

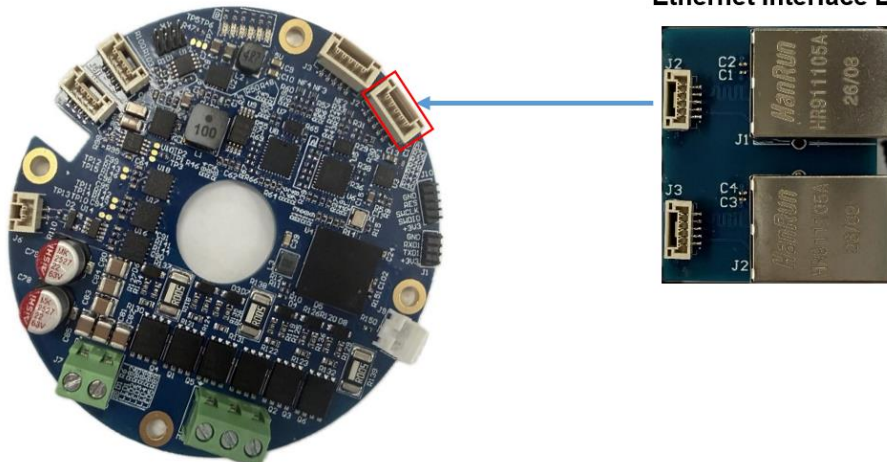
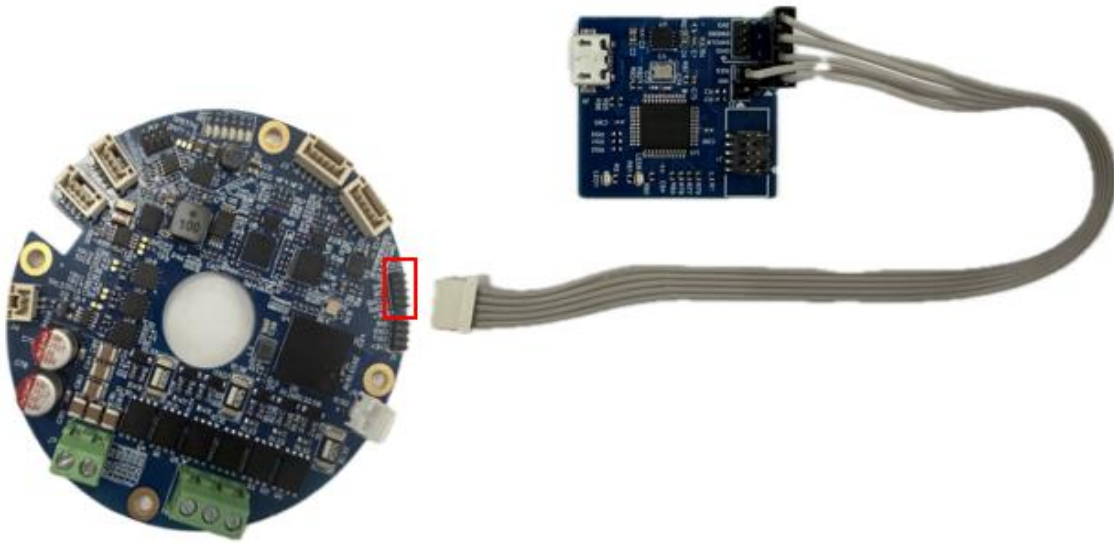


Figure 1-3 Using communication board to connect to host PC

1.3.3.2 RICE

ICE is connected to the Controller board (ARM JTAG connector (5 pins)).



1.3.4 Power Supply

Supply 24V DC power to CN032-9 DC servo solution main control board, and supply 5V from CN032-9 DC servo solution main control board to Tamagawa Encoder and 3.3V to RAA2P3200 encoder board

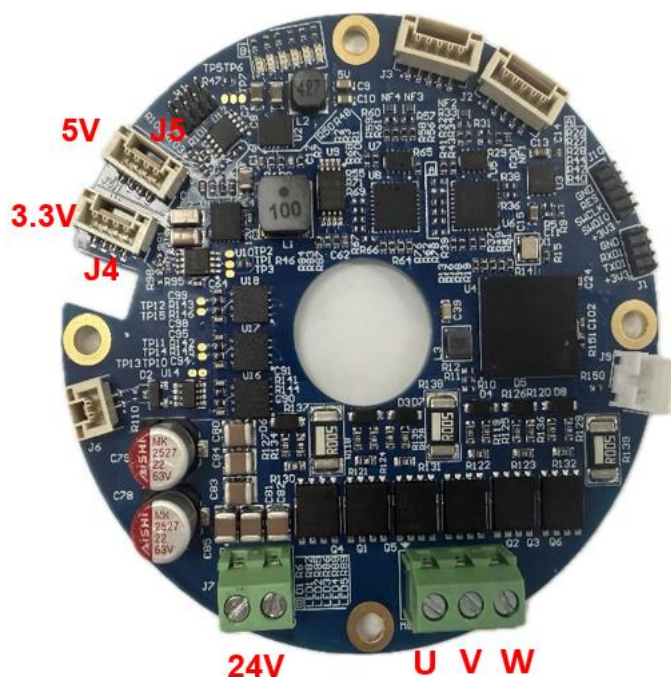


Figure 1-4 CN032-9 DC Servo Solution Main Control Board

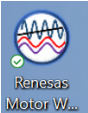
Note: Do not interchange these two encoder connectors. Incorrect connection may cause damage to the ports

2. Operating Environment

This solution kit is for the environment below.

From this chapter onwards, the procedure for operating the DC servo motor control system with 24V DC power supply is described.

Table 2-1 Operating Environment

Item		Description	
		CN032-9 DC Servo Solution (RA8T2)	
CPU	Series	RA8T2 Arm Cortex®-R85	
	Package	R7KA8T2LFLC: 224-pin LFBGA	
Operating frequency		Up to 1GHz	
Operating voltage		3.3V	
Power source for Motor board		24V DC	
Communication protocol		Renesas Motor Workbench 	EtherCAT 
Integrated development environment		Manufactured by Renesas e² studio 2025-04. R20250411-0917 later	
Flexible Support Package (FSP)		RA8T2 FSP6.1.0	
Emulator		Manufactured by SEGGER J-Link made by SST	
Board		Motor board (soldered with RA8T2)	
PC Software		Manufactured by Renesas Renesas Motor Workbench 3.1.2	

3. CN032-9 DC Servo Solution Execution Procedure

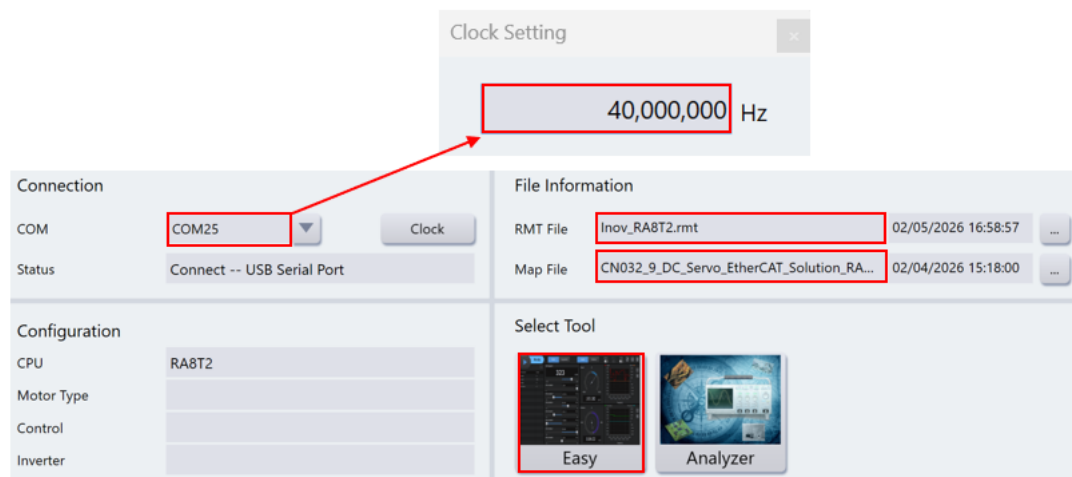
3.1 DC Servo Solution Starting

- ① Connect between the communication board and J1 of motor board
- ② Connect the Tamagawa encoder which contributed from motor to the J5 of motor board
- ③ Connect the RAA2P3200 encoder board J1 to motor board J4
- ④ Connect 24V to DC servo motor board and power on

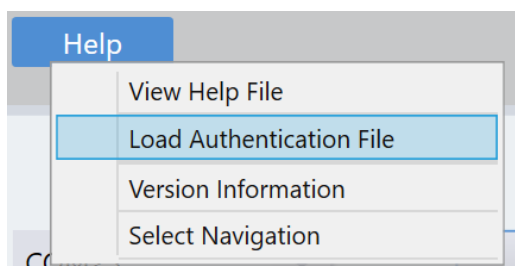
3.2 Renesas Motor Workbench Execution

3.2.1 Installation

- ① Copy the following ZIP file to any folder (*) and unzip the zip file.
Renesas_Motor_Workbench_V312.zip
- ② Open Installer folder and double click renesas_motor_workbench_v312.msi.
- ③ Please follow the installation guide to complete the installation.
- ④ After power-on, double-click Renesas Motor Workbench icon  , select the COM number show in COM item, Input RMT file and Map file. Setting Clock as 40,000,000 Hz



- ⑤ Input renesas_motor_workbench.atf when select [Help] ->[Load Authentication File]



- ⑥ Select Easy Tool to enter the demonstration mode

3.2.2 Demonstration Mode

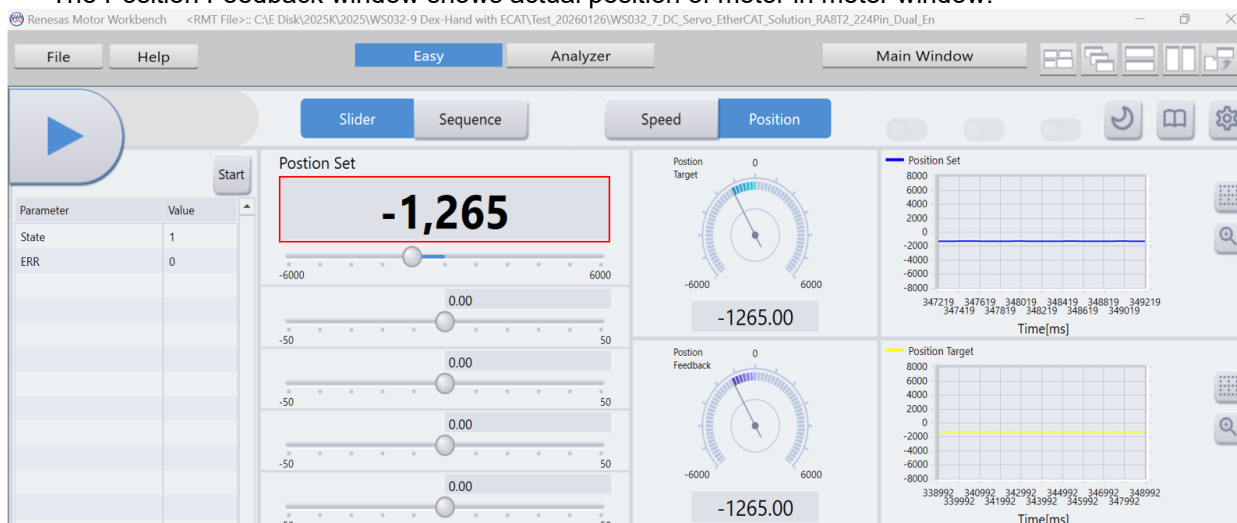
3.2.2.1 Speed Control Mode

- ① Set the speed in "SpeedSet" box, then click RUN button, please also click "Start"
In SpdRef window shows the number of speed set.
In SpeedFdb window shows the actual motor speed read from encoder.
In SpeedSet waveform window, shows the waveform of speed set
In SpeedFeedBack waveform window, shows the waveform of actual motor speed



3.2.2.2 Position Control Mode

- ① Click Position button to shift the control mode from “Speed Control Mode” to “Position Control Mode”.
- ② Set the position number in Position Set box, then click “RUN” button.
- ③ The Position Target window shows position set value in meter window.
The Position Feedback window shows actual position of motor in meter window.



3.3 Renesas Motor Workbench Execution

3.3.1 Preparing EtherCAT Communication

This chapter describes a couple of steps required before actually running the EtherCAT sample project on RA8T2.

3.3.2 TwinCAT®3 Installation

The TwinCAT®3 installation is straight forward and simple. You can download TwinCAT®3 from

<http://www.beckhoff.com/english/download/tc3-downloads.htm?id=1905053019883865>

Please keep in mind that screen shots and operation method may change without notice for newer versions of the software. Above URL will lead you to a screen as shown in Figure 3-1.

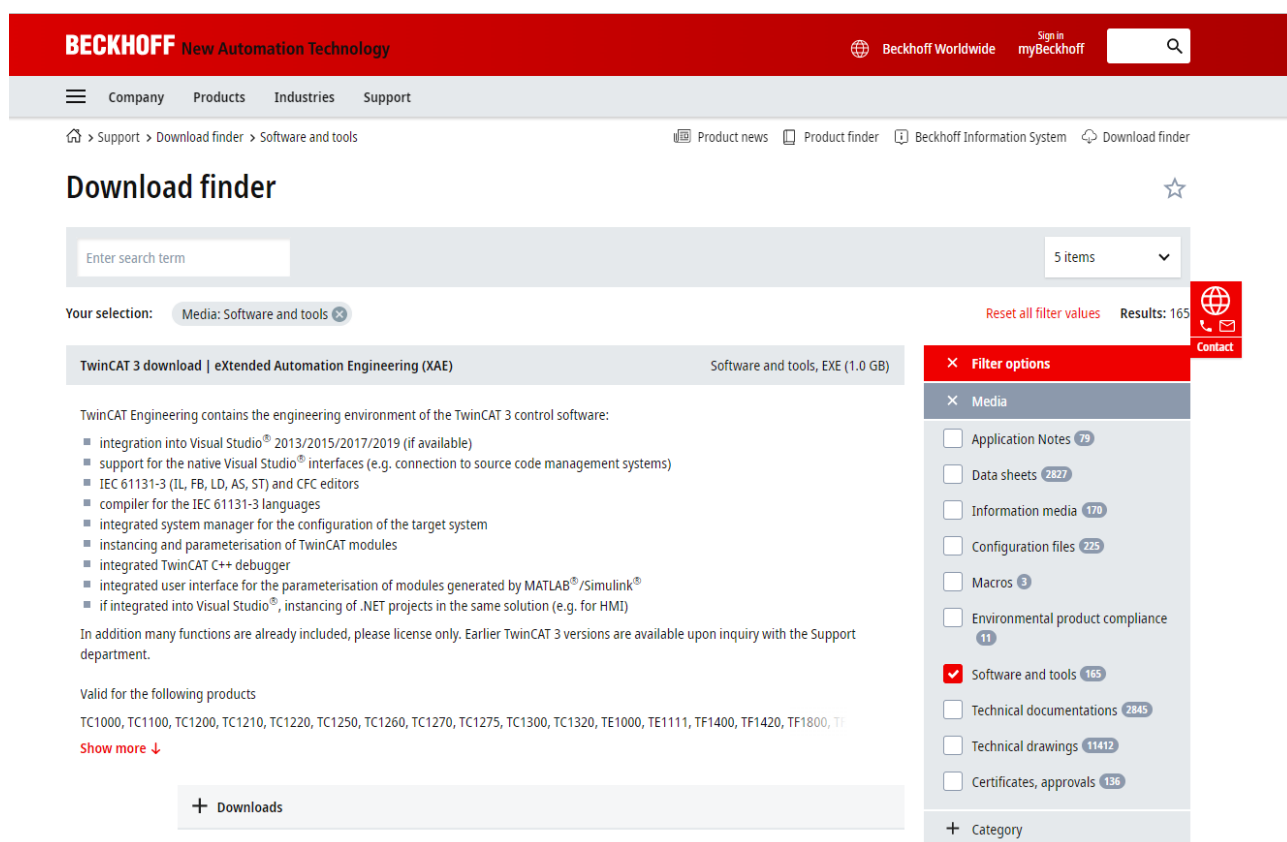


Figure 3-1 TwinCAT®3 Download Screen

Select the TE1xxx|Engineering software package for download. If you are no registered user, you can register and download as guest; the registration is free of charge. You will then get an e mail with a download link. After download, TwinCAT®3 can be installed on your PC. The installation procedure guides you through a number on steps that we will not detail here; simply follow the default installation.

- The installation procedure asks for a serial number. This field can be left empty.
- In the 'Select Installation Level' dialog select TwinCAT PLC – IEC 61131-3 PLC system
- In the 'Select Installation Type' dialog select 30 days demo version (the functions that are required to run our example last longer than 30 days)
- Install all features
- Target directory for the installation should be C:\TwinCAT

After installation one important point needs to be checked in order to control whether the installation has been completed properly. It is the selection of the network adapter for TwinCAT®3. To check that, start TwinCAT®3 and in the start-up screen select “Show Realtime Ethernet Compatible Devices” from the “TwinCAT” menu (see Figure 3-2).

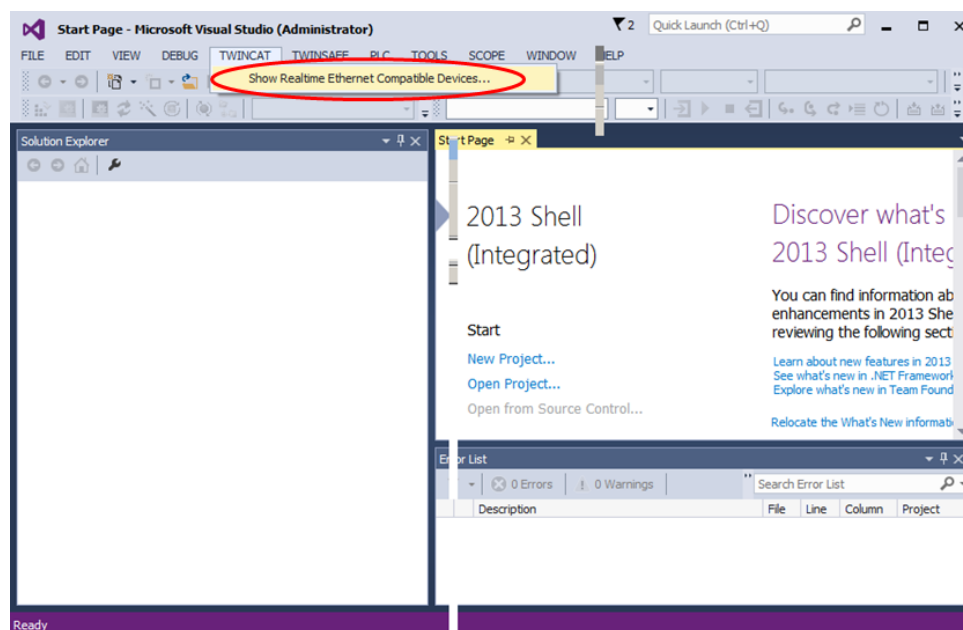


Figure 3-2 Checking Available Network Adapters for TwinCAT®3

TwinCAT®3 will then show you a list of the network adapters in your PC, grouped into several categories. There should be at least one adapter listed under “Compatible devices” as shown in Figure 3-2. Select one of the compatible devices and click “Install”. The network adaptor is then moved to the “Installed and ready to use devices (real-time capable)” category.

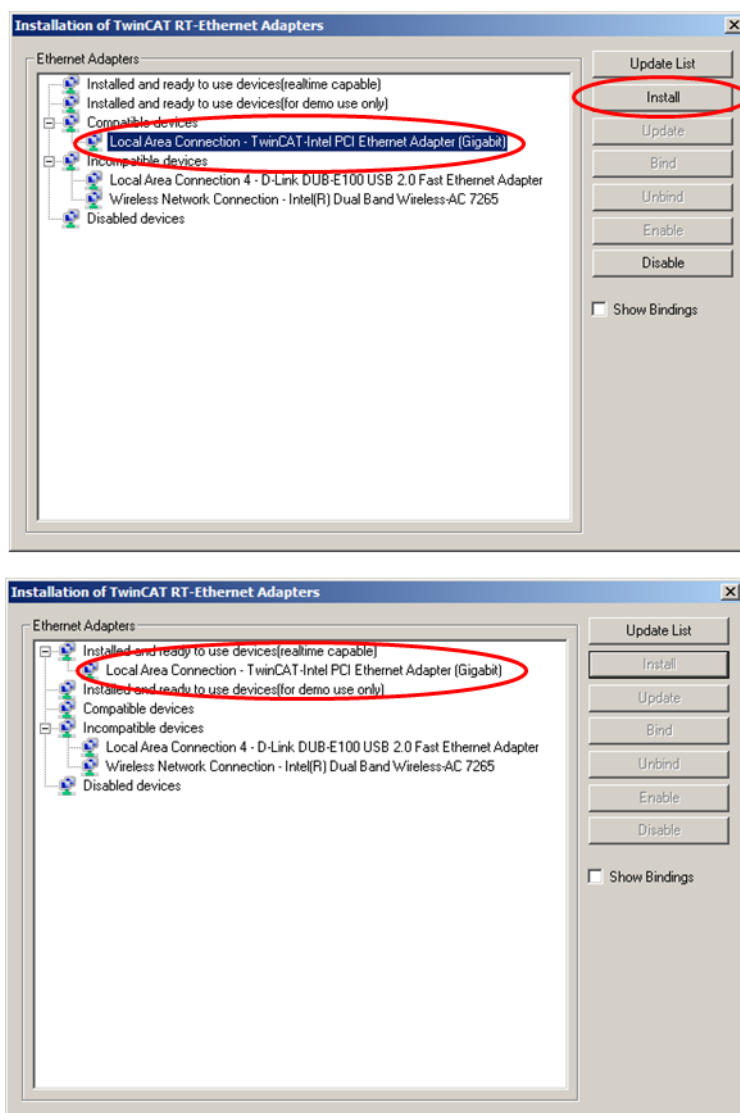


Figure 3-3 Selecting a Compatible Network Adapter

Check the properties of the network adapter that you use for the EtherCAT connection. Open the Network and Sharing Center in MS Windows, right-click on the network adapter used for EtherCAT and select "Properties". You will then see a list of supported protocols and/or services for this adapter like Figure 3-4.

Please uncheck all items that are not EtherCAT related as shown in the screenshot. Then retry to scan for devices.

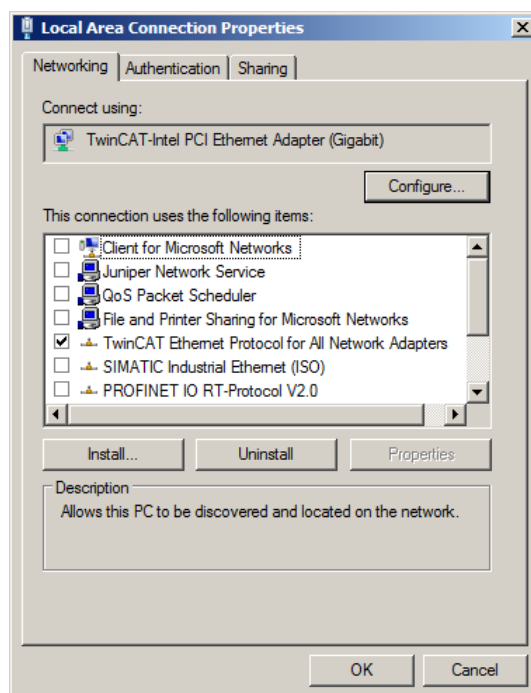


Figure 3-4 Setting the Network Adapter Used for EtherCAT

3.3.3 Copying the ESI File

Obtain the ESI (EtherCAT Slave Information) file "Renesas EtherCAT RA8T2 CiA402.xml" file in "Tool" folder.

Then, copy the obtained file to the following folder, in which TwinCAT®3 has been installed:

"\\TwinCAT\\3.x\\Config\\Io\\EtherCAT"

3.3.4 Confirm Communication with TwinCAT®3

3.3.4.1 Starting TwinCAT®3

- (1) Start the "TwinCAT XAE" program by one of the following methods.
 - a. From the task tray, select [TwinCAT Config Mode] > [TwinCAT XAE (VS2013)]
 - b. From the start menu, select [All Programs] > [Beckhoff] > [TwinCAT 3] > [TwinCAT XAE (VS2013)]
- (2) After starting the program, select [New TwinCAT Project] as shown in Figure 3-5 and create a new project of type TwinCAT XAE Project. Note that the creation of the new project may take several 10 seconds and watch the progress bar in the lower right corner of the TwinCAT®3 window.

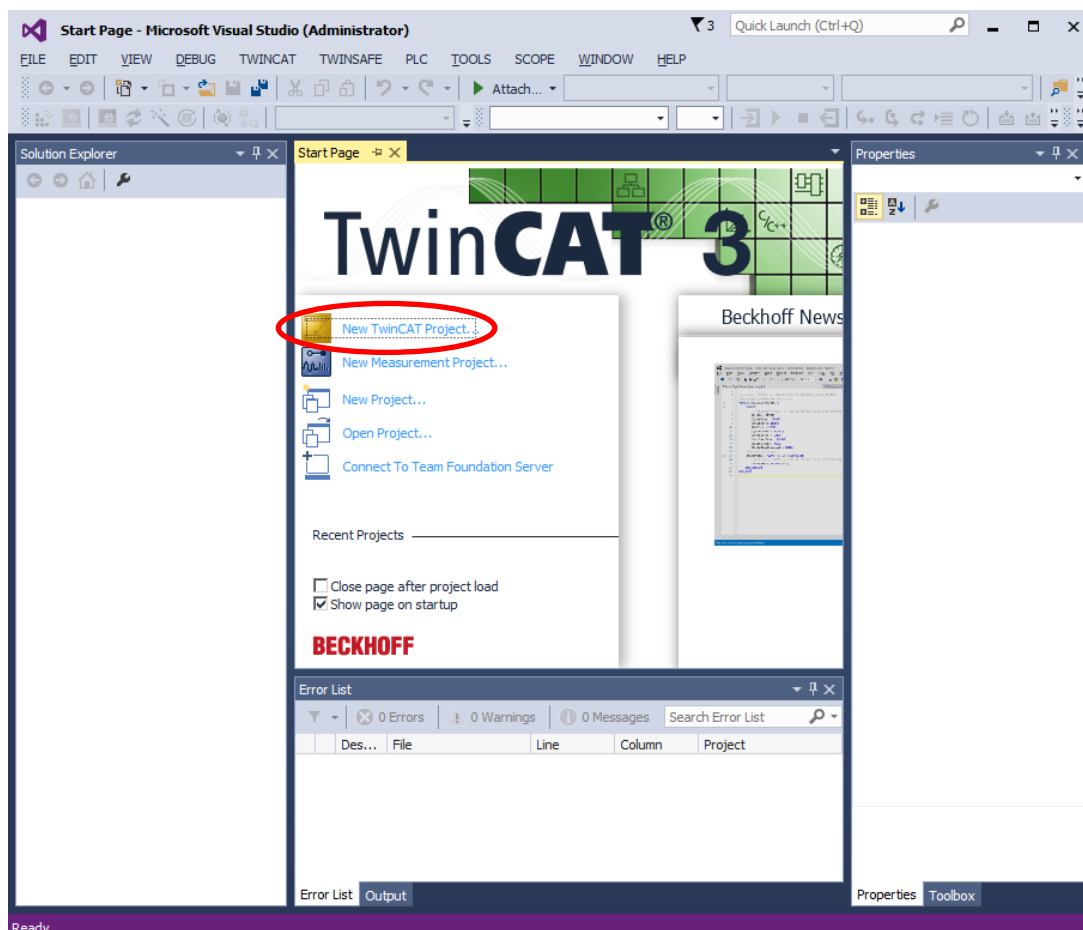


Figure 3-5 TwinCAT®3 Start-up Screen

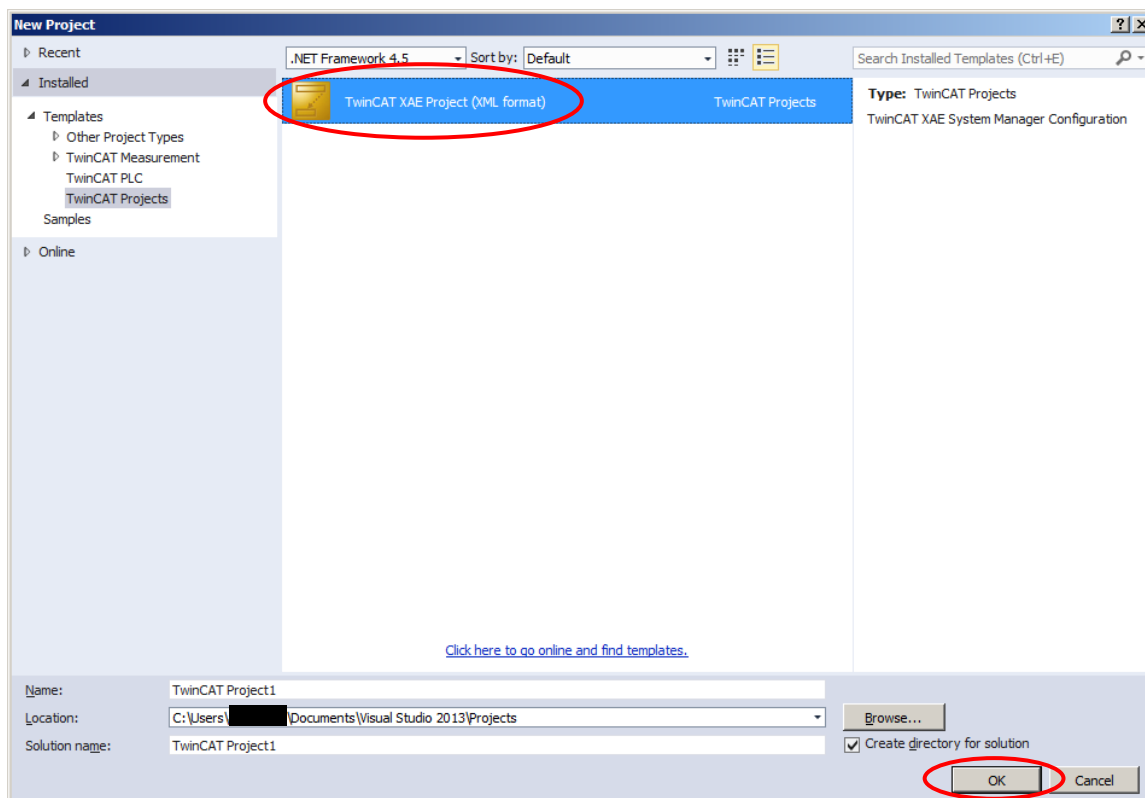


Figure 3-6 Creating a New Project

When the project is created, open the “TWINCAT” menu, chose “EtherCAT Devices” and execute “Reload Device Descriptions”. This ensures, that the device description (aka .ESI file), that you have copied into the TwinCAT®3 installation file structure in chapter 3.3.3 can really be used in our project.

Make sure that the CN032-9 DC servo solution firmware project on the RA8T2 is up and running before the next steps in TwinCAT®3. When the project is not up and running.

(3) Right-click [Devices] and select [Scan] (Figure 3-7)

After you have clicked [Scan], you will be prompted that not all devices in the network can be found automatically. Click ok. TwinCAT®3 will then show you a list of available devices in the network. Select the [EtherCAT] device connected to the network controller that you use for the CN032-9 DC Servo Solution board (see Figure 3-7) and click ok

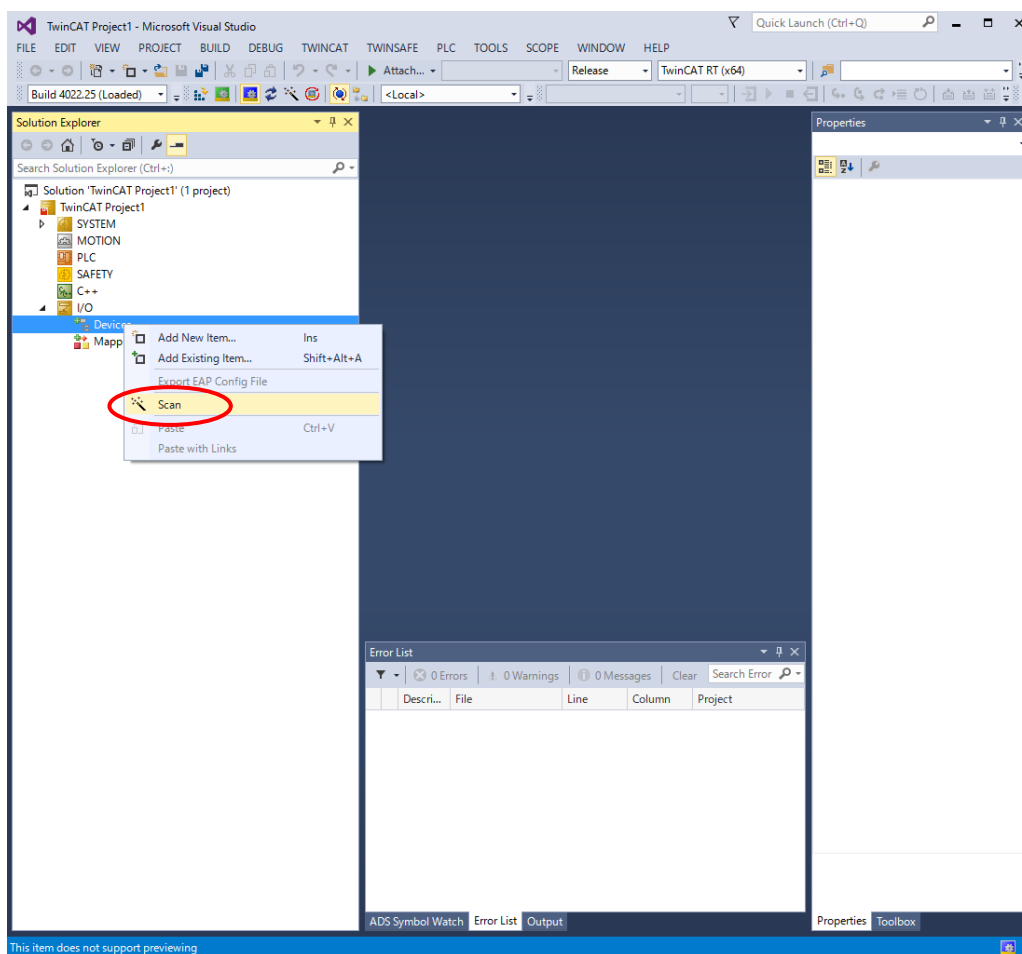


Figure 3-7 Scanning the Network for New Devices

(4) Select only [EtherCAT] and click [OK].

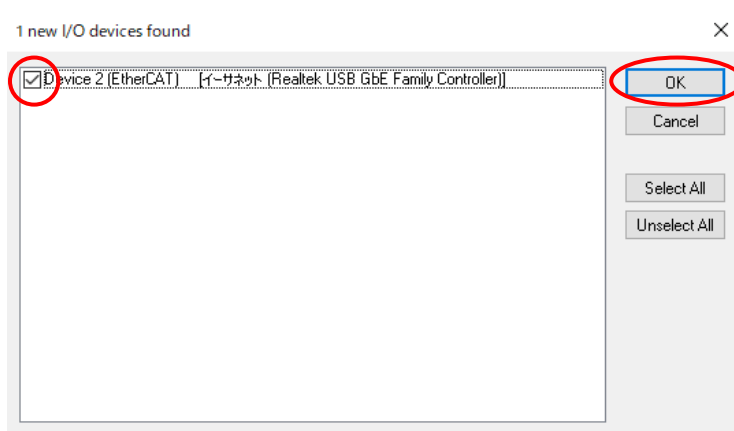
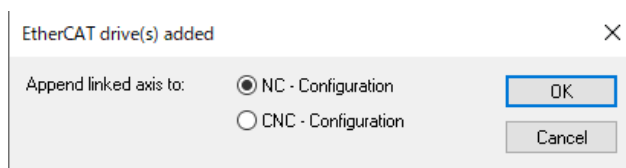


Figure 3-8 Selecting the Proper EtherCAT Device

Note). When scanning the device, the CiA402's ESI is used, so the axis configuration settings are displayed.
Select [NC-Configuration]



After clicking ok you will be prompted, whether you want to “scan for boxes” and to “activate Free Run”. In both cases click “yes”. When Free Run was activated, the RUN and ERR LEDs begin flashing and the L/A IN LED is flashing rapidly.

Note that if you repeat the “Scan” process in the same project in the same environment, you will be informed that no new devices were found. This is no bad thing as there has been no change in the network.

(5) The scan result is displayed as "Drive x (Renesas EtherCAT RA8T2 CiA402 2port)".

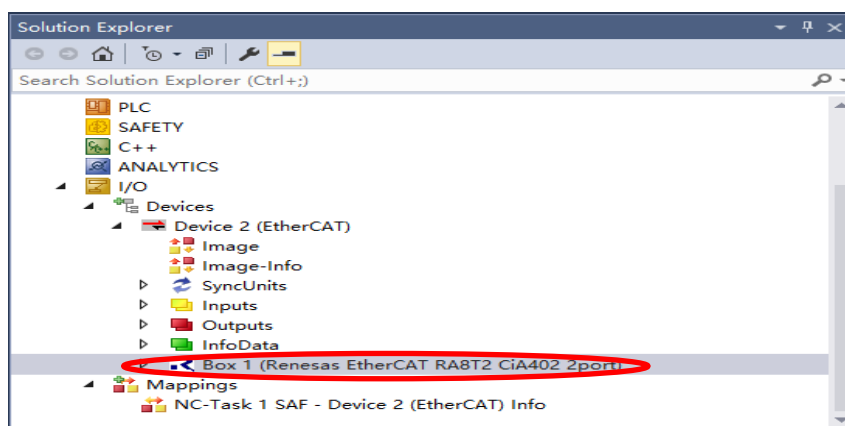


Figure 3-9 Result of Scan for EtherCAT Slave Devices

When the scan result is displayed as "Box 1 (Pxxxxxxx Rxxxxxxx)" as like Figure 3-10 EEPROM data should be written according to chapter 3.3.6. After EEPROM write, Box 1 will be displayed as "Drive x (Renesas EtherCAT RA8T2 CiA402 2port)". EEPROM rewrite is described in chapter 3.3.6.

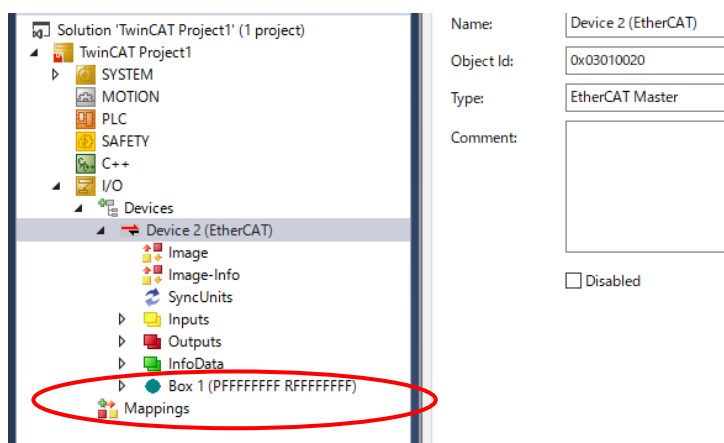


Figure 3-10 Result of Scan for Unknow Devices

Your TwinCAT®3 project is now ready for operation.

3.3.4.2 Checking the Communication Status

To confirm that TwinCAT®3 and the software on the RA8T2 are running properly, it is recommended to check the communication status in TwinCAT®3. For the check, double click the "Box 1 (Renesas EtherCAT RA8T2 CiA402 2port)", select the "Online" tab (2) and check, whether the current state is "OP" (3).

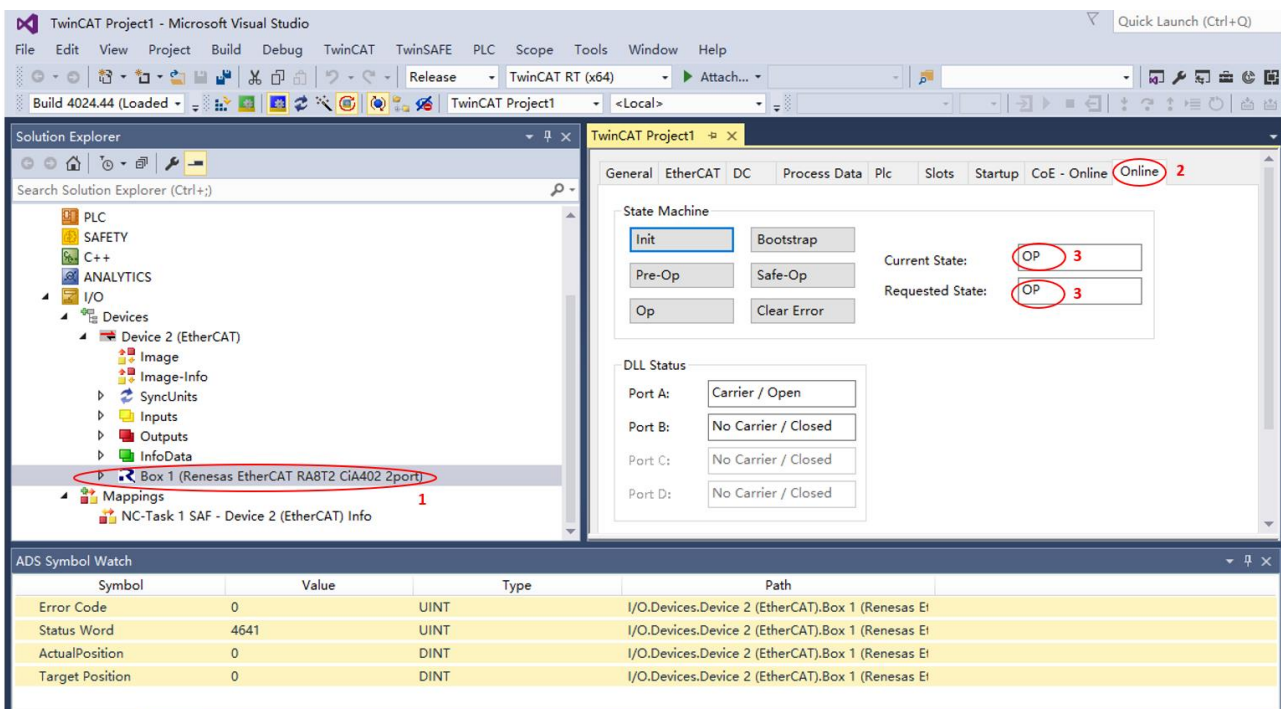


Figure 3-11 Checking the Communication Status

In case that the current state is not "OP" (i.e. "INIT") please restart establishing the communication with the [Restart TwinCAT (Config Mode)] button shown in Figure 3-12. You will then be prompted for

- Restarting TwinCAT System in Config Mode
- Loading I/O Devices
- and Activating Free Run

In all three dialogues simply click "yes". When communication is re-established, the device state in the "Online" tab must change to "OP" as shown in Figure 3-11.

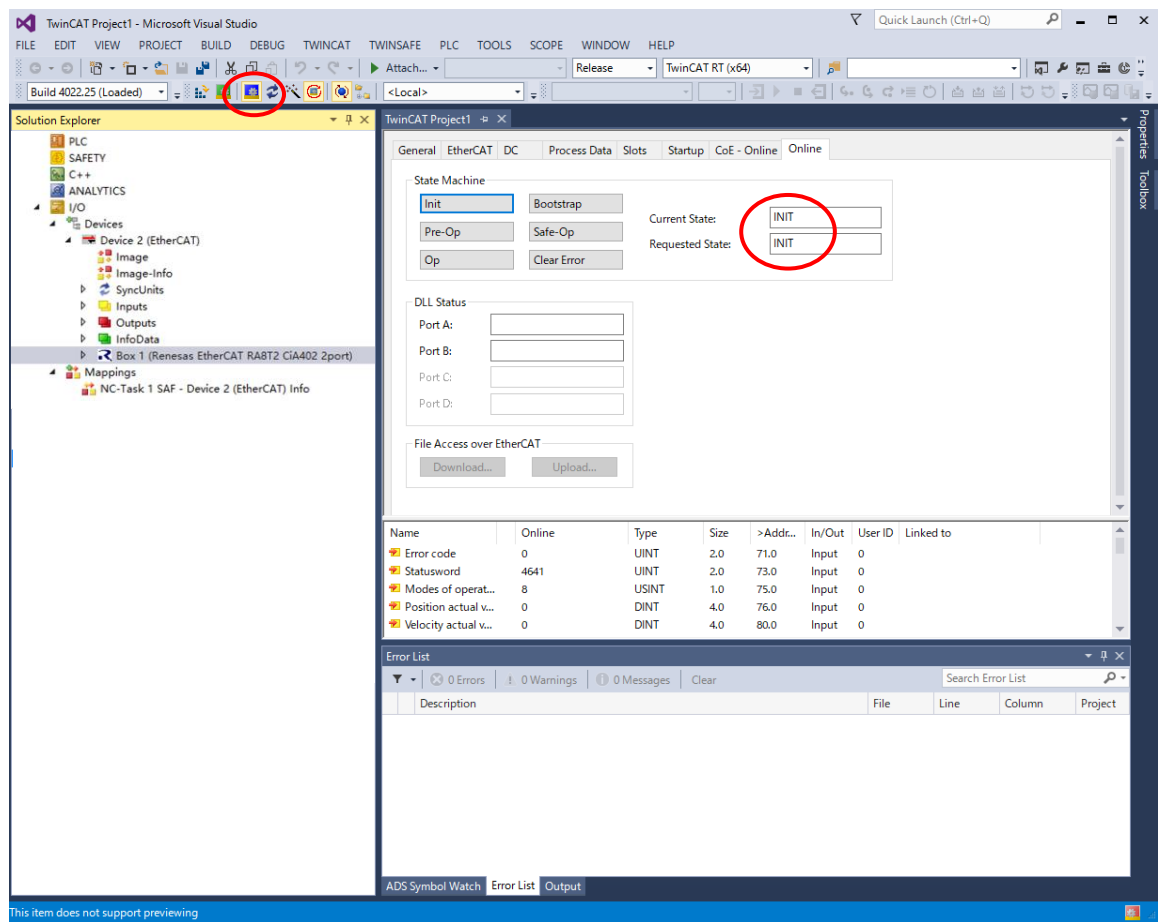


Figure 3-12 Restarting TwinCAT®3 System in Config Mode

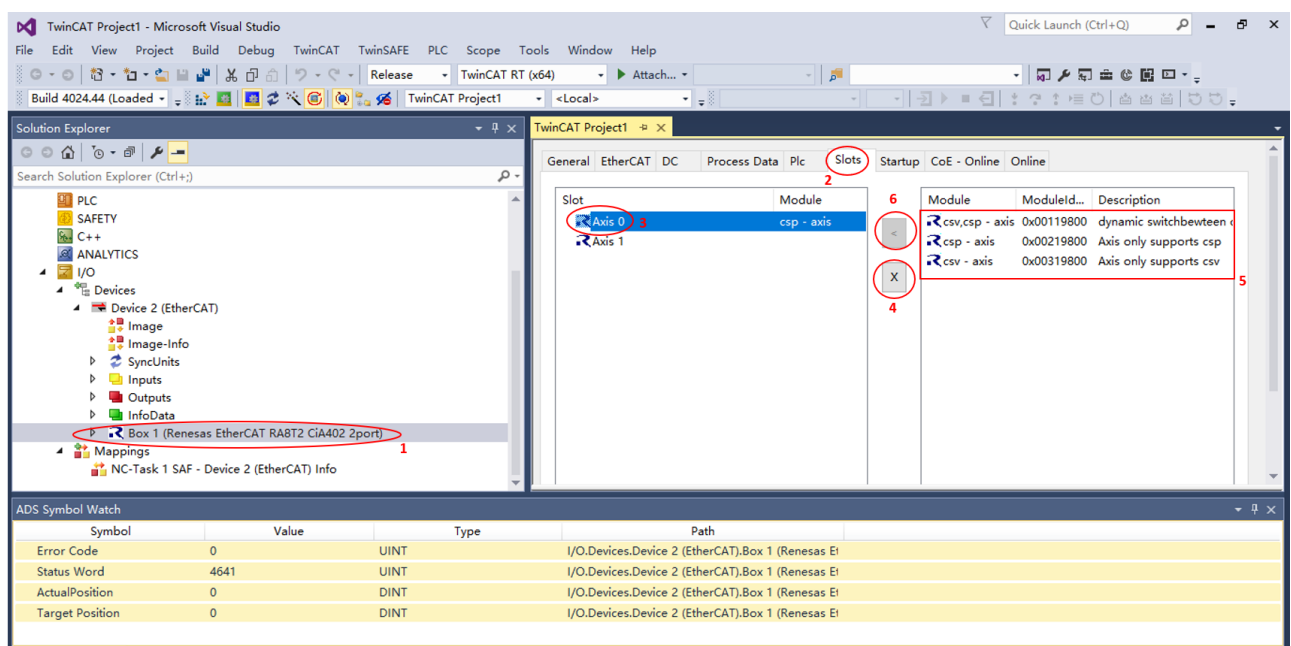
3.3.5 CiA402 Drive Profile Operation

The operation mode of CiA402 drive profile supported by CN032-9 DC servo solution is shown as below operation mode.

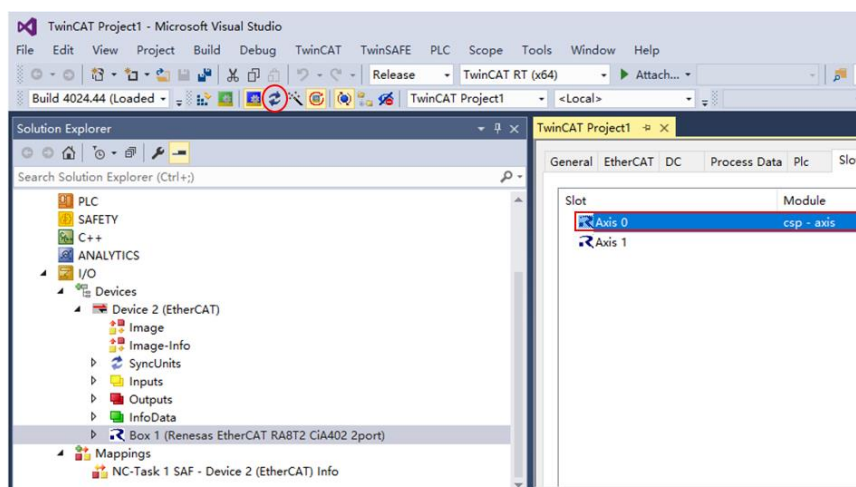
3.3.5.1 Operation mode setting

When changing the operation mode, follow the steps below.

- (1) Select "Drive x (Renesas EtherCAT RA8T2 CiA402 2port)".
- (2) Select "Slots" tab.
- (3) Select "Axis 0".
- (4) Push "X" button to remove current module setting.
- (5) Select any module from multiple modules supported CN032-9 DC Servo Solution.
- (6) Push "<" button to apply selected module setting.

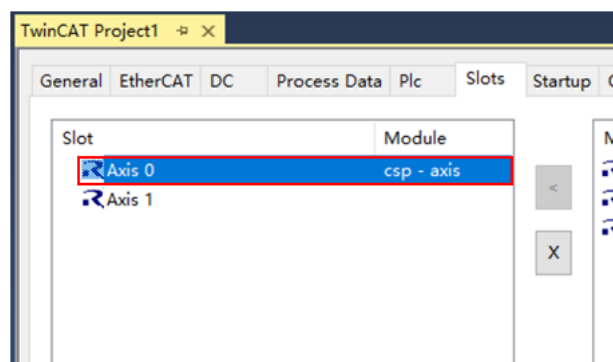


Check the module of Axis 0 was changed to module you selected and then push "↺" button to reload the device.

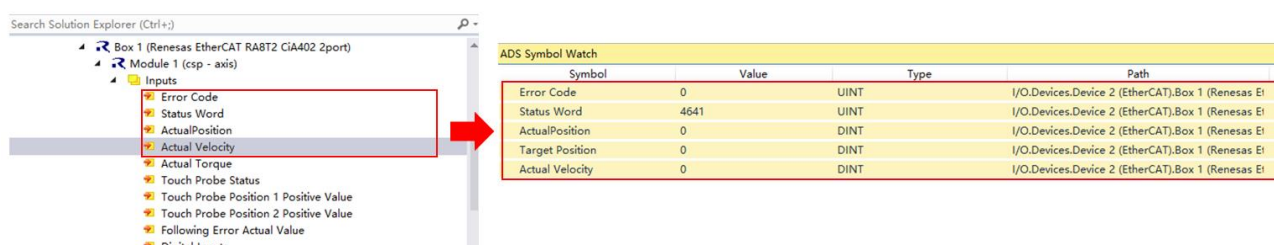


3.3.5.2 Cyclic Synchronous Position Mode (csp)

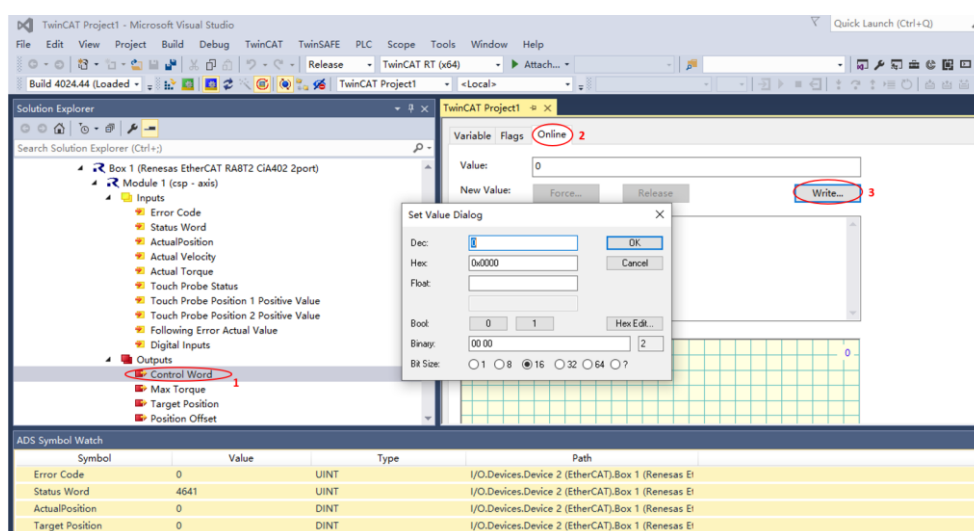
Confirm that the operation mode is “csp”. Otherwise, change the operation mode to “csp” by following 3.3.5.1 Operation mode setting.



Just like the steps described for pp mode, add the input values to an extra watch window for the sake of comfort.



Follow steps (1) to (3) to open a dialog to send the value of Controlword.



Follow steps A) to B) to send the Controlword value and make the motor controllable.

A) Confirm that Statusword is "4641".

ADS Symbol Watch			
Symbol	Value	Type	Path
Error Code	0	UINT	I/O.Devices.Device 2 (EtherCAT).Box 1 (Renesas EtherCAT RA8T2 CiA402 2port).Module 1 (csv - axis).Inputs
Status Word	4641	UINT	I/O.Devices.Device 2 (EtherCAT).Box 1 (Renesas EtherCAT RA8T2 CiA402 2port).Module 1 (csv - axis).Inputs
ActualPosition	0	DINT	I/O.Devices.Device 2 (EtherCAT).Box 1 (Renesas EtherCAT RA8T2 CiA402 2port).Module 1 (csv - axis).Inputs
Target Position	0	DINT	I/O.Devices.Device 2 (EtherCAT).Box 1 (Renesas EtherCAT RA8T2 CiA402 2port).Module 1 (csv - axis).Inputs

B) Set Controlword to "15" and then Statusword will be "4663" (Servo on).

ADS Symbol Watch			
Symbol	Value	Type	Path
Error Code	0	UINT	I/O.Devices.Device 2 (EtherCAT).Box 1 (Renesas EtherCAT RA8T2 CiA402 2port).Module 1 (csv - axis).Inputs
Status Word	4663	UINT	I/O.Devices.Device 2 (EtherCAT).Box 1 (Renesas EtherCAT RA8T2 CiA402 2port).Module 1 (csv - axis).Inputs
Actual Velocity	0	DINT	I/O.Devices.Device 2 (EtherCAT).Box 1 (Renesas EtherCAT RA8T2 CiA402 2port).Module 1 (csv - axis).Inputs
ActualPosition	0	DINT	I/O.Devices.Device 2 (EtherCAT).Box 1 (Renesas EtherCAT RA8T2 CiA402 2port).Module 1 (csv - axis).Inputs

Follow steps (1) to (3) to open a dialog to send the value of Target Position.

Enter the Target Position value and closing the dialog with "Ok" (4) sends the value via EtherCAT.

The screenshot shows the TwinCAT Project1 interface in Microsoft Visual Studio. The Solution Explorer on the left shows the project structure, with 'Target Position' (1) selected under 'Outputs'. The ADS Symbol Watch at the bottom shows the current values: Error Code (0), Status Word (4663), ActualPosition (0), and Target Position (0). The 'Set Value Dialog' is open, showing the 'Dec' field with the value '80000' and the 'OK' button (4) highlighted. The 'Write...' button (3) is also visible in the 'New Value' section. The 'Online' tab (2) is selected in the top right corner.

Symbol	Value	Type	Path
Error Code	0	UINT	I/O.Devices.Device 2 (EtherCAT).Box 1 (Renesas EtherCAT RA8T2 CiA402 2port).Module 1 (csv - axis).Inputs
Status Word	4663	UINT	I/O.Devices.Device 2 (EtherCAT).Box 1 (Renesas EtherCAT RA8T2 CiA402 2port).Module 1 (csv - axis).Inputs
ActualPosition	0	DINT	I/O.Devices.Device 2 (EtherCAT).Box 1 (Renesas EtherCAT RA8T2 CiA402 2port).Module 1 (csv - axis).Inputs
Target Position	0	DINT	I/O.Devices.Device 2 (EtherCAT).Box 1 (Renesas EtherCAT RA8T2 CiA402 2port).Module 1 (csv - axis).Inputs

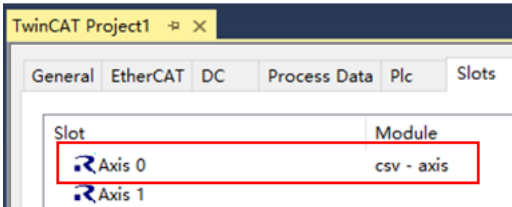
Then, the motor rotates until the Position actual value counts up to the Target Position value.

ADS Symbol Watch			
Symbol	Value	Type	Path
Error Code	0	UINT	I/O.Devices.Device 2 (EtherCAT).Box 1 (Renesas E
Status Word	4663	UINT	I/O.Devices.Device 2 (EtherCAT).Box 1 (Renesas E
ActualPosition	2812	DINT	I/O.Devices.Device 2 (EtherCAT).Box 1 (Renesas E
Target Position	80000	DINT	I/O.Devices.Device 2 (EtherCAT).Box 1 (Renesas E
ADS Symbol Watch Error List			

When stop the motor control, set Controlword to “6” and then Statusword will be “4641”.

3.3.5.3 Cyclic Synchronous Velocity Mode (csv)

Confirm that the operation mode is “csv”. Otherwise, change the operation mode to “csv” by following 3.3.5.1 Operation mode setting.



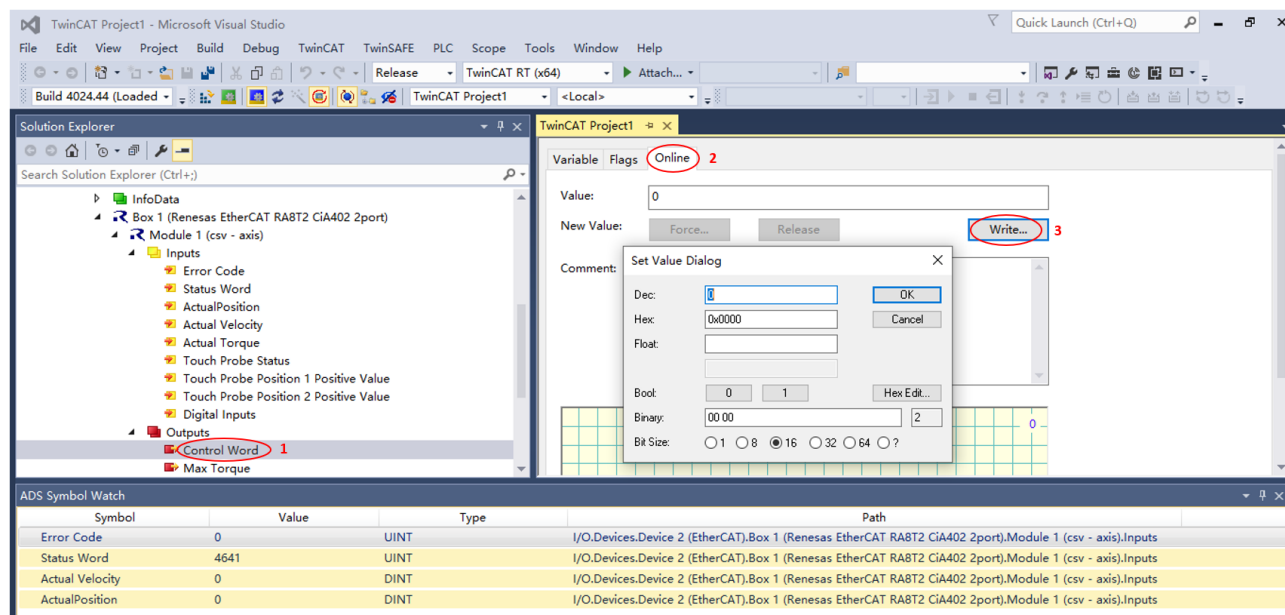
Just like the steps described for pp mode, add the input values to an extra watch window for the sake of comfort.

- Error Code
- Status Word
- ActualPosition
- Actual Velocity
- Actual Torque
- Touch Probe Status
- Touch Probe Position 1 Positive Value
- Touch Probe Position 2 Positive Value
- Digital Inputs

→

ADS Symbol Watch			
Symbol	Value	Type	Path
Error Code	0	UINT	I/O.Devices.Device 2 (EtherCAT).Box 1 (Renesas EtherCAT RA8T2 CIA402 2port).Module 1 (csv - axis).Inputs
Status Word	4641	UINT	I/O.Devices.Device 2 (EtherCAT).Box 1 (Renesas EtherCAT RA8T2 CIA402 2port).Module 1 (csv - axis).Inputs
Actual Velocity	0	DINT	I/O.Devices.Device 2 (EtherCAT).Box 1 (Renesas EtherCAT RA8T2 CIA402 2port).Module 1 (csv - axis).Inputs
ActualPosition	0	DINT	I/O.Devices.Device 2 (EtherCAT).Box 1 (Renesas EtherCAT RA8T2 CIA402 2port).Module 1 (csv - axis).Inputs

Follow steps (1) to (3) to open a dialog to send the value of Controlword.



Follow steps A) to B) to send the Controlword value and make the motor controllable.

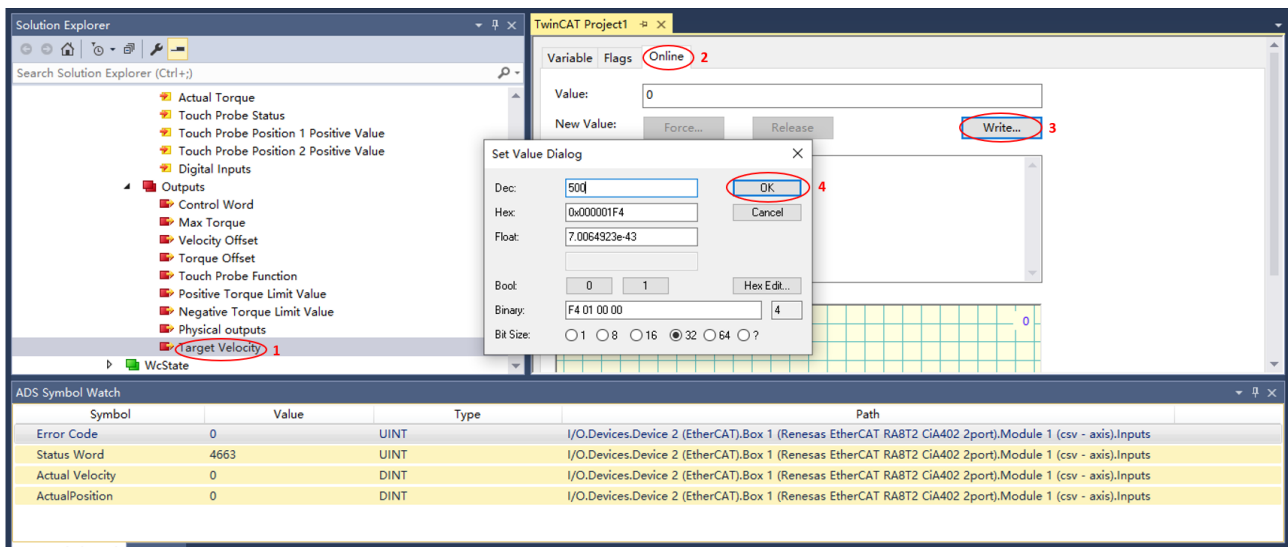
A) Confirm that Statusword is “4641”.

Symbol	Value	Type	Path
Error Code	0	UINT	I/O.Devices.Device 2 (EtherCAT).Box 1 (Renesas EtherCAT RA8T2 CiA402 2port).Module 1 (csv - axis).Inputs
Status Word	4641	UINT	I/O.Devices.Device 2 (EtherCAT).Box 1 (Renesas EtherCAT RA8T2 CiA402 2port).Module 1 (csv - axis).Inputs
Actual Velocity	0	DINT	I/O.Devices.Device 2 (EtherCAT).Box 1 (Renesas EtherCAT RA8T2 CiA402 2port).Module 1 (csv - axis).Inputs
ActualPosition	0	DINT	I/O.Devices.Device 2 (EtherCAT).Box 1 (Renesas EtherCAT RA8T2 CiA402 2port).Module 1 (csv - axis).Inputs

B) Set Controlword to “15” and then Statusword will be “4663” (Servo on).

Symbol	Value	Type	Path
Error Code	0	UINT	I/O.Devices.Device 2 (EtherCAT).Box 1 (Renesas EtherCAT RA8T2 CiA402 2port).Module 1 (csv - axis).Inputs
Status Word	4663	UINT	I/O.Devices.Device 2 (EtherCAT).Box 1 (Renesas EtherCAT RA8T2 CiA402 2port).Module 1 (csv - axis).Inputs
Actual Velocity	0	DINT	I/O.Devices.Device 2 (EtherCAT).Box 1 (Renesas EtherCAT RA8T2 CiA402 2port).Module 1 (csv - axis).Inputs
ActualPosition	0	DINT	I/O.Devices.Device 2 (EtherCAT).Box 1 (Renesas EtherCAT RA8T2 CiA402 2port).Module 1 (csv - axis).Inputs

Follow steps (1) to (3) to open a dialog to send the value of Target Velocity.
Enter the Target Velocity value and closing the dialog with “Ok” (4) sends the value via EtherCAT.



Then, the motor rotates while controlling the Velocity actual value to the Target Velocity value.

Symbol	Value	Type	Path
Error Code	0	UINT	I/O.Devices.Device 2 (EtherCAT).Box 1 (Renesas EtherCAT RA8T2 CiA402 2port).Module 1 (csv - axis).Inputs
Status Word	4663	UINT	I/O.Devices.Device 2 (EtherCAT).Box 1 (Renesas EtherCAT RA8T2 CiA402 2port).Module 1 (csv - axis).Inputs
Actual Velocity	512	DINT	I/O.Devices.Device 2 (EtherCAT).Box 1 (Renesas EtherCAT RA8T2 CiA402 2port).Module 1 (csv - axis).Inputs
ActualPosition	0	DINT	I/O.Devices.Device 2 (EtherCAT).Box 1 (Renesas EtherCAT RA8T2 CiA402 2port).Module 1 (csv - axis).Inputs

When stop the motor control, set Controlword to “6” and then Statusword will be “4641”.

You can now modify the parameters as you like for further experiments. The full “command set” that is covered by the Controlword and the specific meaning of error codes or Statusword can be studied in the “CANopen Device Profile Drives and Motion Control” control document, that is available from the CAN in Automation organization at

<https://www.can-cia.org/can-knowledge/canopen/cia402/>

Please note that the document can only be downloaded by organization members. Copies of old versions are vagabonding on the www.

3.3.6 EEPROM Data Update on CN032-9 DC Servo Solution

If the link between TwinCAT®3 and CN032-9 DC Servo Solution is established, you can update EEPROM data on the kit from TwinCAT®3. The EEPROM contains identification data like VendorID or Product ID.

The EEPROM is blank when purchasing the CN032-9 DC Servo Solution, in this case Box1 will be displayed as “PFFFFFFFF RFFFFFFFF” as shown in Figure 3-12. Depending on the history of your board you may as well find other data in the EEPROM; therefore, we recommend updating the EEPROM in any case as described in the next steps.

To update the EEPROM, double-click on Box 1 (1), select the “EtherCAT” tab (2) and click the “Advanced settings” button (3) as illustrated in Figure 3-13.

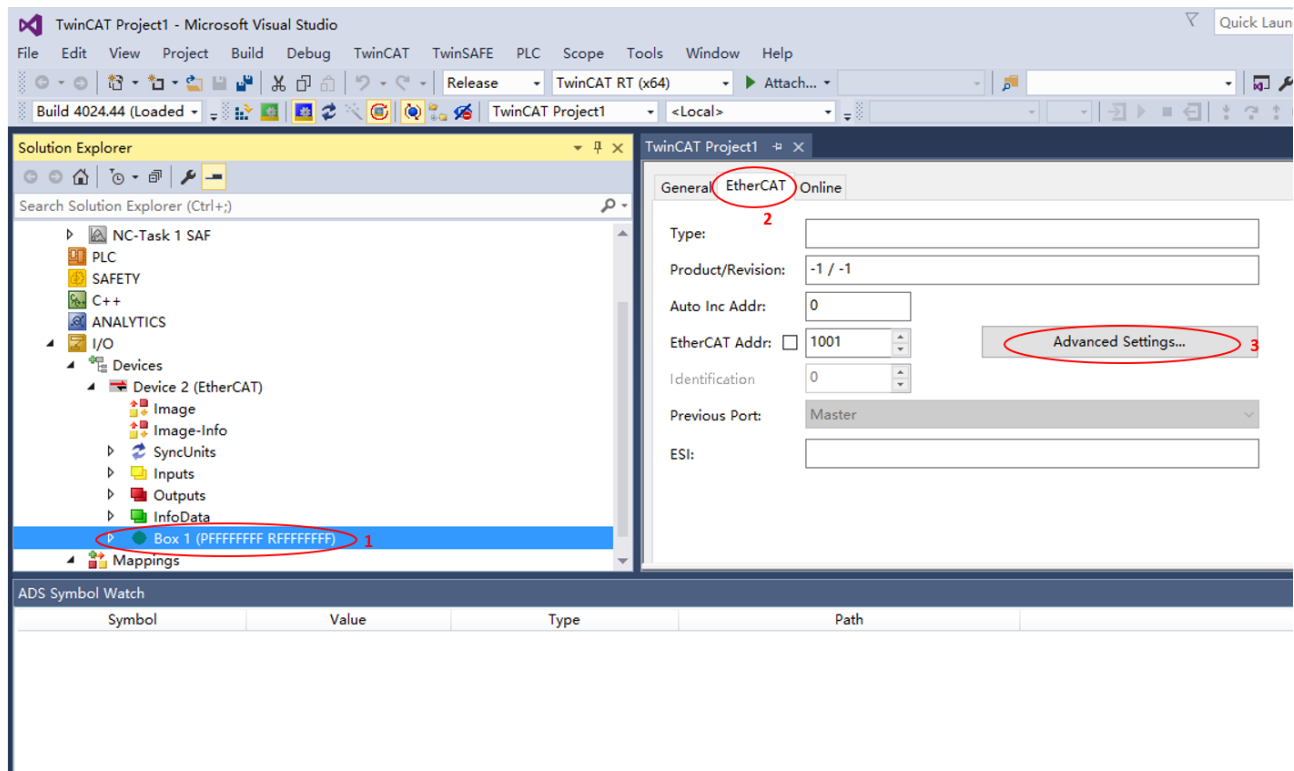


Figure 3-13 Updating the EEPROM

Then expand the list of advanced settings accordingly, so that you can select the “Hex Editor” (1). In the “Hex Editor” view click “Download from List” (2) (See Figure 3-12). This list contains numerous devices for which the required EEPROM content is ready to use.

At the bottom of the list you find as well some Renesas devices. Select “Renesas EtherCAT RA8T2 CiA402 2port” and click ok (see Figure 3-14).

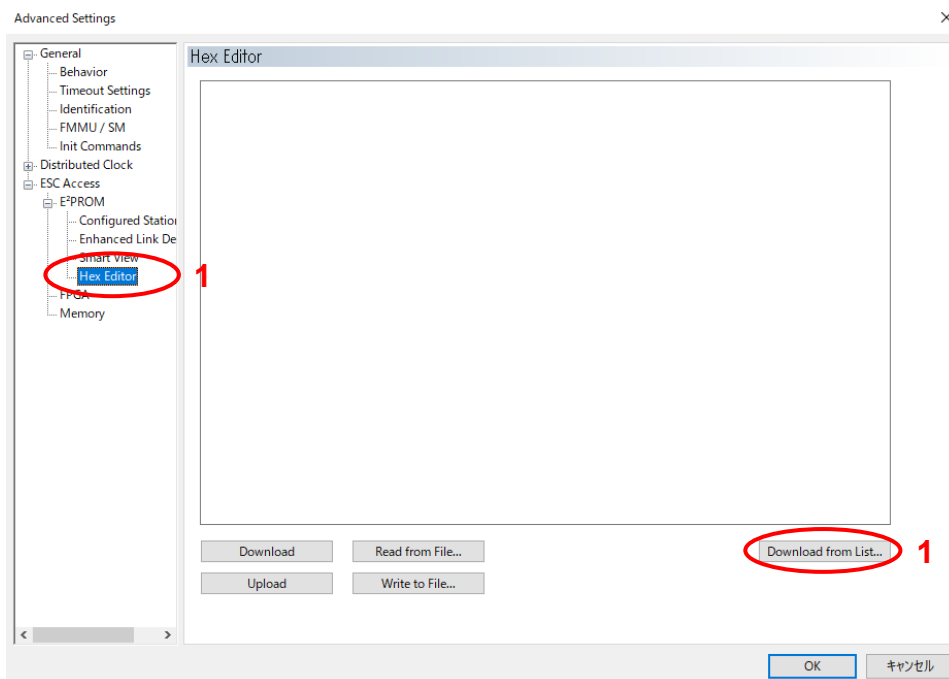


Figure 3-14 Selecting the Hex Editor for EEPROM File Download

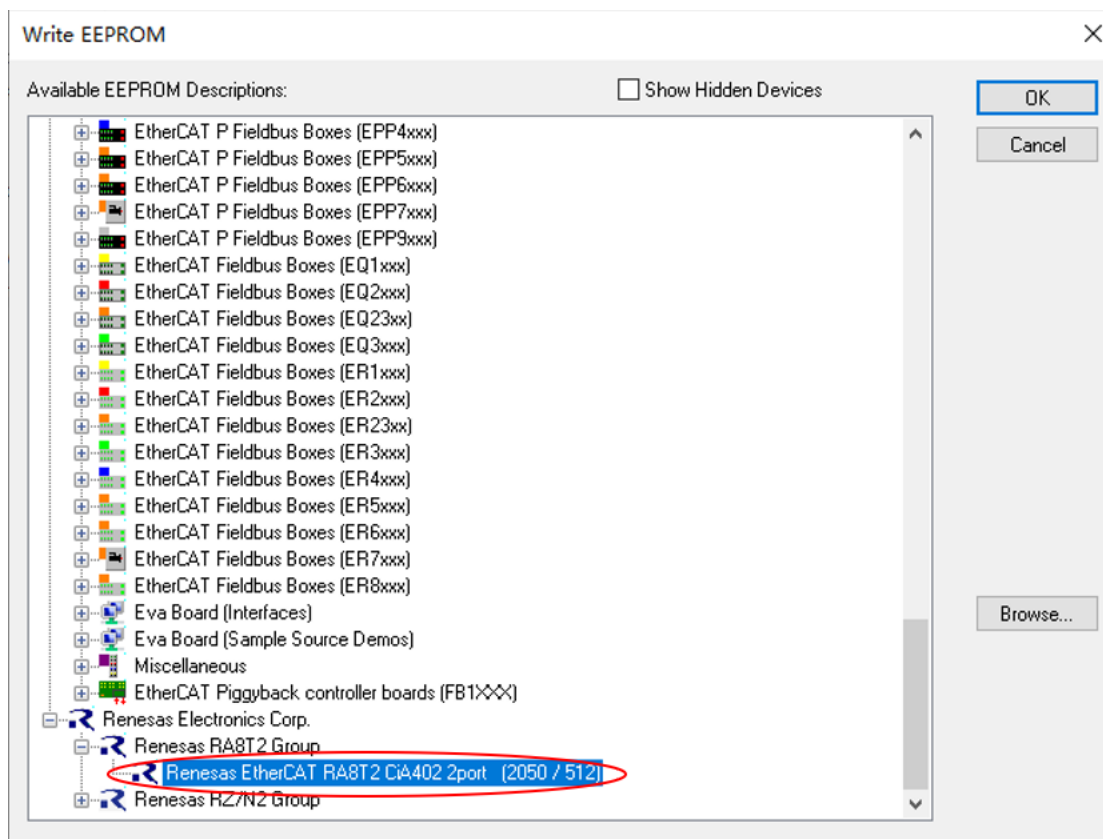


Figure 3-15 Selecting the Proper EEPROM Description File

The data is then downloaded to the EEPROM; at the end of the download process the data is automatically verified. With “Upload” (see Figure 3-14) you can check, whether the EEPROM programming was successful.

Now repeat the device scan shown in Figure 3-13. TwinCAT®3 will now scan once more for devices and it will find the Renesas EtherCAT RA8T2 CiA402 2port Solution board with the updated description data in EEPROM. The subsequent “Check Configuration” dialog shows you the list of found items. Copy the “Renesa EtherCAT RA8T2 CiA402 2port” device with “Copy All” into the list of “Configured Items” (1). When the representation of the devices in the list changes to green, leave the dialog with “OK” (2).

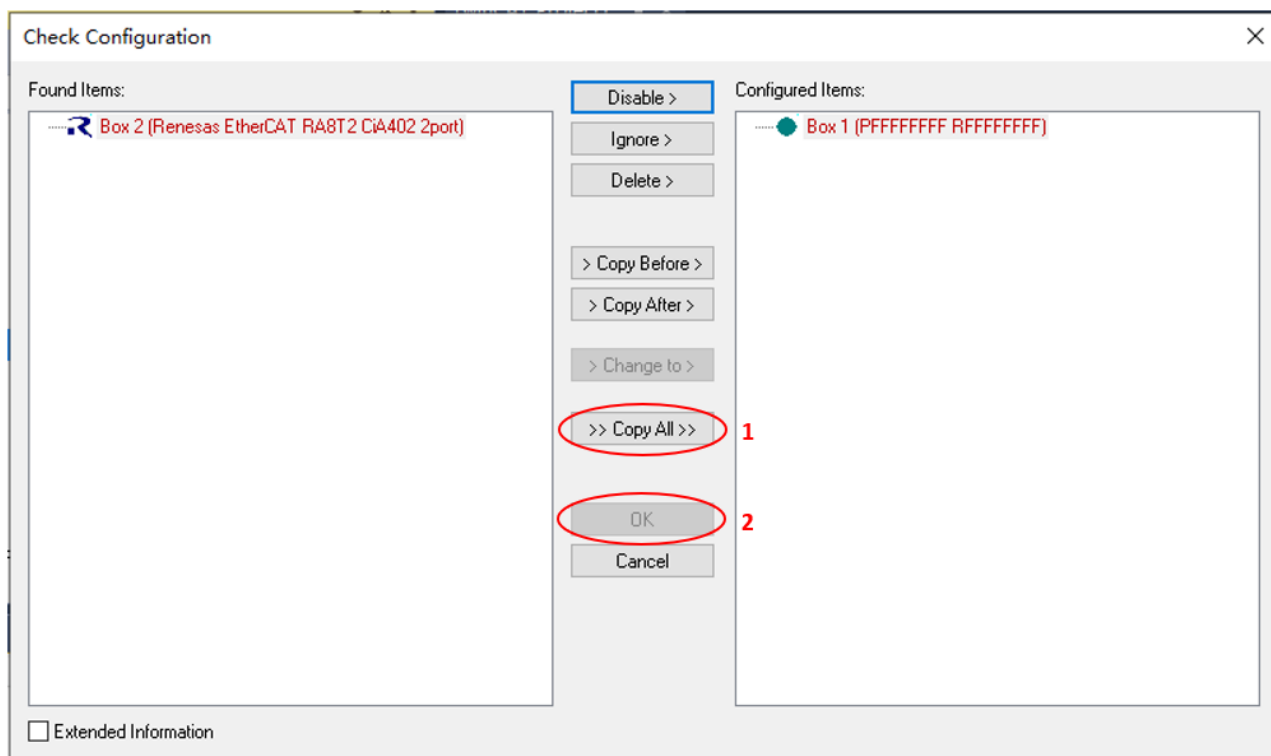


Figure 3-16 Copy the found device into the configuration

4. Reference Documents

[RA8T2 - 1GHz Arm Cortex-M85 Microcontroller Optimized for Motor Control Applications | Renesas](#)

Technical Updates/Technical News

(The latest information can be downloaded from the Renesas Electronics Website.)

Website and Support

Renesas Electronics Website

<http://www.renesas.com/>

Inquiries

<http://www.renesas.com/contact/>

Revision History

Rev.	Date	Description	
		Page	Summary
1.00	Mar. 31, 2026	—	First edition issued

General Precautions in the Handling of Micro processing Unit and Microcontroller Unit Products

The following usage notes are applicable to all Micro processing unit and Microcontroller unit products from Renesas. For detailed usage notes on the products covered by this document, refer to the relevant sections of the document as well as any technical updates that have been issued for the products.

1. Precaution against Electrostatic Discharge (ESD)

A strong electrical field, when exposed to a CMOS device, can cause destruction of the gate oxide and ultimately degrade the device operation. Steps must be taken to stop the generation of static electricity as much as possible, and quickly dissipate it when it occurs. Environmental control must be adequate. When it is dry, a humidifier should be used. This is recommended to avoid using insulators that can easily build up static electricity. Semiconductor devices must be stored and transported in an anti-static container, static shielding bag or conductive material. All test and measurement tools including work benches and floors must be grounded. The operator must also be grounded using a wrist strap. Semiconductor devices must not be touched with bare hands. Similar precautions must be taken for printed circuit boards with mounted semiconductor devices.

2. Processing at power-on

The state of the product is undefined at the time when power is supplied. The states of internal circuits in the LSI are indeterminate and the states of register settings and pins are undefined at the time when power is supplied. In a finished product where the reset signal is applied to the external reset pin, the states of pins are not guaranteed from the time when power is supplied until the reset process is completed. In a similar way, the states of pins in a product that is reset by an on-chip power-on reset function are not guaranteed from the time when power is supplied until the power reaches the level at which resetting is specified.

3. Input of signal during power-off state

Do not input signals or an I/O pull-up power supply while the device is powered off. The current injection that results from input of such a signal or I/O pull-up power supply may cause malfunction and the abnormal current that passes in the device at this time may cause degradation of internal elements. Follow the guideline for input signal during power-off state as described in your product documentation.

4. Handling of unused pins

Handle unused pins in accordance with the directions given under handling of unused pins in the manual. The input pins of CMOS products are generally in the high-impedance state. In operation with an unused pin in the open-circuit state, extra electromagnetic noise is induced near the LSI, an associated shoot-through current flows internally, and malfunctions occur due to the false recognition of the pin state as an input signal become possible.

5. Clock signals

After applying a reset, only release the reset line after the operating clock signal becomes stable. When switching the clock signal during program execution, wait until the target clock signal is stabilized. When the clock signal is generated with an external resonator or from an external oscillator during a reset, ensure that the reset line is only released after full stabilization of the clock signal. Additionally, when switching to a clock signal produced with an external resonator or by an external oscillator while program execution is in progress, wait until the target clock signal is stable.

6. Voltage application waveform at input pin

Waveform distortion due to input noise or a reflected wave may cause malfunction. If the input of the CMOS device stays in the area between V_{IL} (Max.) and V_{IH} (Min.) due to noise, for example, the device may malfunction. Take care to prevent chattering noise from entering the device when the input level is fixed, and also in the transition period when the input level passes through the area between V_{IL} (Max.) and V_{IH} (Min.).

7. Prohibition of access to reserved addresses

Access to reserved addresses is prohibited. The reserved addresses are provided for possible future expansion of functions. Do not access these addresses as the correct operation of the LSI is not guaranteed.

8. Differences between products

Before changing from one product to another, for example to a product with a different part number, confirm that the change will not lead to problems. The characteristics of a micro processing unit or microcontroller unit products in the same group but having a different part number might differ in terms of internal memory capacity, layout pattern, and other factors, which can affect the ranges of electrical characteristics, such as characteristic values, operating margins, immunity to noise, and amount of radiated noise. When changing to a product with a different part number, implement a system-evaluation test for the given product

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